

EXHIBIT 13



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/729,298	06/03/2015	Emre Alptekin	FIS920150066US1	2837
133964 7590 04/21/2016 HOFFMAN WARNICK LLC 540 Broadway 4th Floor ALBANY, NY 12207			EXAMINER NGUYEN, LAWRENCE-LIN T	
			ART UNIT 2897	PAPER NUMBER
			NOTIFICATION DATE 04/21/2016	DELIVERY MODE ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

PTOCommunications@hoffmanwarnick.com

Office Action Summary	Application No. 14/729,298	Applicant(s) ALPTEKIN ET AL.	
	Examiner LAWRENCE-LINH T. NGUYEN	Art Unit 2897	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 2 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1) ☒ Responsive to communication(s) filed on 02/26/2016.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a) ☐ This action is **FINAL**. 2b) ☐ This action is non-final.

3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☒ Claim(s) 1-20 is/are pending in the application.
5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☐ Claim(s) _____ is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☒ Claim(s) 1-20 are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

10) ☐ The specification is objected to by the Examiner.

11) ☐ The drawing(s) filed on _____ is/are: a) ☐ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some** c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☐ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. _____.

4) ☐ Other: _____.

DETAILED ACTION

Election/Restrictions

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.
2. Restriction to one of the following inventions is required under 35 U.S.C. 121:
 - I. Claims **1-14**, drawn to **a method of**, classified in **438** subclass **648**.
 - II. Claims **15-20**, drawn to **a device of**, classified in **257** subclass **757**.
3. The inventions are independent or distinct, each from the other because of the following reasons:
 - a. Inventions Group I and Group II are related as process of making and product made. The inventions are distinct if either or both of the following can be shown: (1) that the process as claimed can be used to make another and materially different product or (2) that the product as claimed can be made by another and materially different process (MPEP § 806.05(f)).
 - b. In the instant case One aspect of the present invention involves depositing a first liner layer in a contact opening such that an intermix region is formed at an interface of the bottom portion of the contact opening and an upper portion of the semiconductor substrate thereunder, e.g., source and drain. Unreacted (non-diffused) portions of the first liner layer may be removed prior to completion of

fabricating the contact. Therefore, aspects of the present invention do not increase the deposited metal thickness in the contact openings.

4. **Applicant is advised that the reply to this requirement to be complete must include:**

(i) an election of a species to be examined even though the requirement may be traversed (37 CFR 1.143) **and**

(ii) identification of the claims encompassing the elected species or grouping of patentably indistinct species, including any claims subsequently added. An argument that a claim is allowable or that all claims are generic is considered nonresponsive unless accompanied by an election.

5. The election may be made with or without traverse. To preserve a right to petition, the election must be made with traverse. If the reply does not distinctly and specifically point out supposed errors in the election of species requirement, the election shall be treated as an election without traverse. Traversal must be presented at the time of election in order to be considered timely. Failure to timely traverse the requirement will result in the loss of right to petition under 37 CFR 1.144. If claims are added after the election, applicant must indicate which of these claims are readable on the elected species or grouping of patentably indistinct species.

6. Should applicant traverse on the ground that the species, or groupings of patentably indistinct species from which election is required, are not patentably distinct, applicant should submit evidence or identify such evidence now of record showing them to be obvious variants or clearly admit on the record that this is the case. In either

instance, if the examiner finds one of the species unpatentable over the prior art, the evidence or admission may be used in a rejection under 35 U.S.C. 103 or pre-AIA 35 U.S.C. 103(a) of the other species.

7. The examiner has required restriction between product or apparatus claims and process claims. Where applicant elects claims directed to the product/apparatus, and all product/apparatus claims are subsequently found allowable, withdrawn process claims that include all the limitations of the allowable product/apparatus claims should be considered for rejoinder. All claims directed to a nonelected process invention must include all the limitations of an allowable product/apparatus claim for that process invention to be rejoined.

8. In the event of rejoinder, the requirement for restriction between the product/apparatus claims and the rejoined process claims will be withdrawn, and the rejoined process claims will be fully examined for patentability in accordance with 37 CFR 1.104. Thus, to be allowable, the rejoined claims must meet all criteria for patentability including the requirements of 35 U.S.C. 101, 102, 103 and 112. Until all claims to the elected product/apparatus are found allowable, an otherwise proper restriction requirement between product/apparatus claims and process claims may be maintained. Withdrawn process claims that are not commensurate in scope with an allowable product/apparatus claim will not be rejoined. See MPEP § 821.04.

Additionally, in order for rejoinder to occur, applicant is advised that the process claims should be amended during prosecution to require the limitations of the product/apparatus claims. **Failure to do so may result in no rejoinder.** Further, note

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that the prohibition against double patenting rejections of 35 U.S.C. 121 does not apply where the restriction requirement is withdrawn by the examiner before the patent issues. See MPEP § 804.01.

Conclusion

1. Any inquiry concerning this communication or earlier communications from the examiner should be directed to LAWRENCE-LINH T. NGUYEN whose telephone number is (571) 272-6267. The examiner can normally be reached on M-F (8AM - 4PM EST).
2. If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Fernando Toledo can be reached on (571) 272-1867. The fax phone number for the organization where this application or proceeding is assigned is (571) 272-8300.
3. Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at (866) 217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call (800) 786-9199 (in USA or CANADA) or (571) 272-1000.

/Lawrence-Linh T Nguyen/
Examiner, Art Unit 2897

/Fernando L. Toledo/
Supervisory Patent Examiner, Art
Unit 2897

Application/Control Number: 14/729,298
Art Unit: 2897

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant:	Alptekin et al.	Conf. No.:	2837
Serial No.:	14/729,298	Art Unit:	2897
Filing Date:	06/03/2015	Examiner:	Nguyen, Lawrence-Lin T.
Title:	Contacts to Semiconductor Substrate and Methods of Forming Same	Docket No.:	FIS920150066US1 (GFFT-0376)

Mail Stop Amendment
Commissioner for Patents
PO Box 1450
Alexandria, VA 22313-1450

RESPONSE TO RESTRICTION REQUIREMENT

I. INTRODUCTORY COMMENTS

This paper is being filed in response to the Restriction Requirement dated 21 April 2016.

Applicant hereby elects invention I, claims 1-14, without traverse.

II. REMARKS

Claims 1–20 are pending in this application. The Office requires election between invention I (claims 1-14), drawn to “a method,” and invention II (claims 15-20), drawn to “a device.” (Office Action at page 2).

Applicant hereby elects invention I (claims 1-14) for examination on the merits, without traverse. Claims 15-20 stand withdrawn.

Should the Office require anything further from the Applicant, the Examiner is invited to contact Applicant’s undersigned representative at the number listed below.

Respectfully submitted,

/Amanda J. Smith /
Amanda J. Smith
Reg. No. 71,112

Date: June 20, 2016

Hoffman Warnick LLC
540 Broadway, 4th Floor
Albany, New York 12207
Phone: (518) 449-0044
Fax: (518) 449-0047

Electronic Acknowledgement Receipt

EFS ID:	26109411
Application Number:	14729298
International Application Number:	
Confirmation Number:	2837
Title of Invention:	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME
First Named Inventor/Applicant Name:	Emre Alptekin
Customer Number:	133964
Filer:	Amanda J Smith
Filer Authorized By:	
Attorney Docket Number:	FIS920150066US1
Receipt Date:	20-JUN-2016
Filing Date:	03-JUN-2015
Time Stamp:	11:38:29
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Response to Election / Restriction Filed	FIS920150066US1_ResponseTo RestrictionRequirement.pdf	65761 6f75468b06dd8660e5d14d96353ba16274 344086	no	2

Warnings:

Information:

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/729,298	06/03/2015	Emre Alptekin	FIS920150066US1	2837
133964 7590 09/26/2016 HOFFMAN WARNICK LLC 540 Broadway 4th Floor ALBANY, NY 12207			EXAMINER NGUYEN, LAWRENCE-LIN T	
			ART UNIT 2897	PAPER NUMBER
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PTOCommunications@hoffmanwarnick.com

Detailed Action

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.
2. This non-final Office action is responsive to Applicants' application filed **06/20/2016**. Claims **1-20** are presented for examination and are rejected for the reasons indicated herein below.

Election/Restrictions

3. Applicant's election without traverse of Group I, claims **1-14** in the reply filed on **06/20/2016** is acknowledged.
4. Claims **15-20** are withdrawn pursuant to a restriction requirement.
5. Now claims **1-14** are pending.

Specification Objections

6. In the Specification, page 4, second paragraph, objected to because reference characters "**110**" and "**112**" have both been used to designate "**Substrate**". Appropriate correction is required.

Claim Rejections - 35 USC § 102(a)(1)

7. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

(a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale or otherwise available to the public before the effective filing date of the claimed invention.

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

8. Claims **1-14** are rejected under 35 U.S.C. 102(a1), as being anticipated by

Heinrich et al., (US20120161324A1, hereinafter Heinrich).

9. In reclaim **1**, in FIG. 2, Heinrich discloses a method for forming a contact in a dielectric layer to a semiconductor substrate, the method comprising:

- forming a contact opening(**223**) in the dielectric layer(**221**) over the semiconductor substrate(**201**) to expose an upper portion of the semiconductor substrate(**201**);
- depositing a first liner layer(**205**) to conformally coat the contact opening(**223**);
- causing a portion of the first liner layer(**205**) to diffuse into the upper portion of the semiconductor substrate(**201**) to form a first intermix region(**253**) at the upper portion of the semiconductor substrate(**201**);
- depositing a refractory metal layer(**203**) over the first intermix

region(**YY**); and

- depositing a metal(**224a**) in the contact opening(**223**) thereby forming the contact(**223_Contact**) (See § **[0035-0046]**).

10. In reclaim **2**, in FIG. 2b, Heinrich discloses a method of claim **1**, further comprising: removing a remaining non-diffused portion of the first liner layer(**205**) to expose the first intermix region(**223_Bottom**) at the upper portion of the semiconductor substrate(**201**) prior to depositing the refractory metal layer (See § **[0039]**).

11. In reclaim **3**, in FIG. 2, Heinrich discloses a method of claim **1**, wherein the depositing the first liner layer(**205**) includes depositing at least one of: nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), and ytterbium (Yb) (See § **[0006 0009]**).

12. In reclaim **4**, in FIG. XX, aaa discloses a method of claim 1, wherein depositing the refractory metal layer(**203**) includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN) (See § **[0006]**).

13. In reclaim **5**, in FIG. 2, Heinrich discloses a method of claim 1, wherein depositing the first liner layer includes performing physical vapor deposition (See § **[0006]**).

14. In reclaim **7**, in FIG. 2, Heinrich discloses a method of claim 1, wherein the depositing the first liner layer(**205**) includes depositing nickel platinum (NiPt), the depositing the refractory metal layer(**203**) includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN), and the depositing the metal(**224a**) in the opening includes

depositing tungsten (W) (See § [0045]).

15. In reclaim 8, in FIG. 2, Heinrich discloses a method for forming a set of contacts in a dielectric layer to a field effect transistor, the field effect transistor including a source, a drain, and a gate between the source and the drain, the method comprising:

- forming a contact opening(223) in the dielectric layer(221) to expose an upper-portion of each of the source(252_Source) and the drain(252_Drain);
- depositing a first liner layer(205) to conformally coat each contact opening(223) and causing a portion of the first liner layer(205) to diffuse into the upper portion of each of the source(252_Source) and the drain(252_Drain) to form a first intermix region(253) at the upper portion of the each of the source(252_Source) and the drain(252_Drain);
- depositing a refractory metal layer(203) over the first intermix region(253); and
- depositing a metal(224a) to substantially coat the refractory metal layer(203) and substantially fill each of the openings(223) thereby forming the set of contacts(223_Contact) (See § [0035-0046]).

16. In reclaim 9, in FIG. 2b, Heinrich discloses a method of claim 8, further comprising: removing a remaining non-diffused portion of the first liner layer(205) to expose the intermix region(223_Bottom) prior to depositing the refractory metal layer(203) (See § [0039]).

17. In reclaim **10**, in FIG. 2, Heinrich discloses a method of claim 8, wherein the depositing the first liner layer(**205**) includes depositing nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), and ytterbium (Yb) (*See § [0006 0009]*).

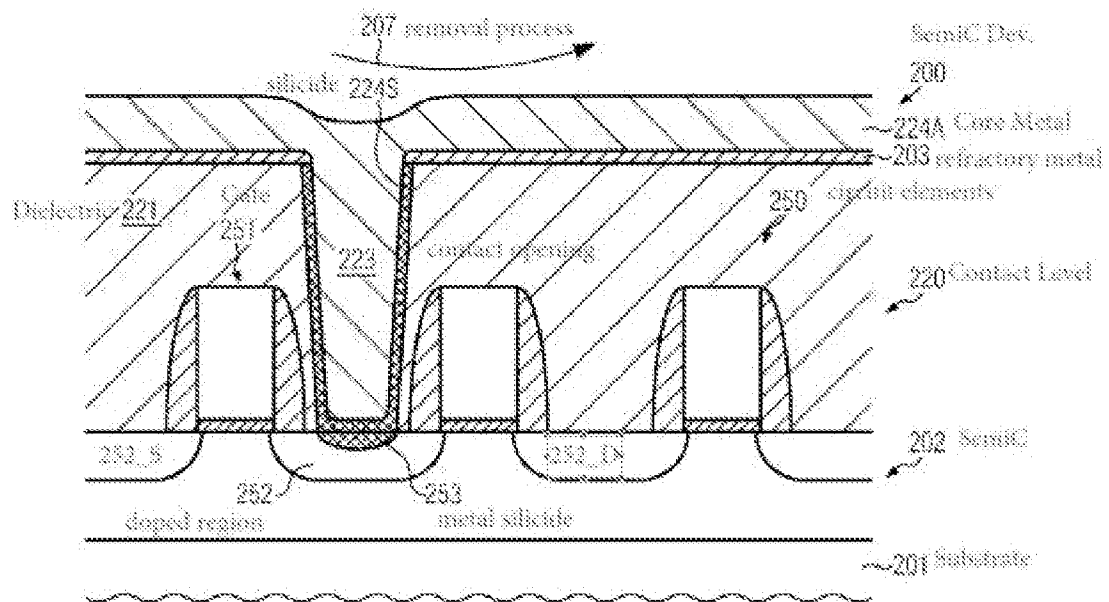
18. In reclaim **11**, in FIG. 2, Heinrich discloses a method of claim 8, wherein the depositing the refractory metal layer(**203**) includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN) (*See § [0006]*).

12. In reclaim **12**, in FIG. 2, Heinrich discloses a method of claim 8, wherein the depositing the first liner layer(**205**) includes performing physical vapor deposition (*See § [0006]*).

19. In reclaim **14**, in FIG. 2, Heinrich discloses a method of claim 13, wherein the depositing the first liner layer(**205**) includes depositing nickel platinum (NiPt), the depositing the refractory metal layer(**203**) includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN), and the depositing the metal(**224a**) in the opening includes depositing tungsten (W) (*See § [0045]*).

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20. (Also see US20120161324A1_FIG2hMod which is reproduced below for convenience).



Claim Rejections - 35 USC § 103

22. The following is a quotation of 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

A patent may not be obtained through the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries set forth in Graham v. John Deere Co., 383 U.S. 1, 148 USPQ 459 (1966), that are applied for establishing a background for determining obviousness under 35 U.S.C. 103(a) are summarized as follows: (See MPEP Ch. 2141)

Determining the scope and contents of the prior art;

Ascertaining the differences between the prior art and the claims in issue; Resolving the level of ordinary skill in the pertinent art; and Evaluating evidence of secondary considerations for indicating obviousness or nonobviousness

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

23. Claims **6 & 13** is rejected under 35 U.S.C. 103(a) as being unpatentable by **Heinrich et al.**, (**US20120161324A1**, hereinafter **Heinrich**) in view of **Hu et al.**, (**US20140027883A1**, hereinafter **Hu**).

24. In reclaim **6**, Heinrich is silent in the teaching of a method of claim 1, further comprising: polishing the metal; and performing a laser anneal of the opening such that a second intermix region is formed at the upper portion of the semiconductor

substrate and the upper surface of semiconductor substrate, wherein the second intermix region is composed of materials used for the first intermix region and a material used for the refractory metal layer.

25. But in the same field of endeavor of invention, Hu teaches a method of claim 1, further comprising: polishing the metal; and performing a laser anneal of the opening such that a second intermix region is formed at the upper portion of the semiconductor substrate and the upper surface of semiconductor substrate, wherein the second intermix region is composed of materials used for the first intermix region and a material used for the refractory metal layer.

26. Hu is evidence that ordinary workers skilled in the art would find reasons, suggestions or motivations to modify the device of Hu with Heinrich. Therefore, at the time the invention was made, it would have been obvious to use the teaching of Hu in the device of Heinrich since the superlattice formed from the first refractory metal nitride material and the second refractory metal nitride material may also improve the thermal stability of the bottom electrode or interconnect (*See § [0016]*).

27. In reclaim **13**, in FIG. 2, Heinrich in view of Hu, Hu teaches a method of claim 8, further comprising: polishing the metal in each of the openings; and performing a laser anneal of each of the openings such that a second intermix region is formed at the upper portion of the respective source and drain thereunder, wherein the second intermix region is composed of materials used for the first intermix region and a material used for the refractory metal layer (*See § [0016 0027]*).

Conclusion

29. Any inquiry concerning this communication or earlier communications from the examiner should be directed to LAWRENCE-LINH T. NGUYEN whose telephone number is (571) 272-6267. The examiner can be reached on M-F (8AM - 4PM EST).

30. If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Fernando Toledo can be reached on (571) 272-1867. The fax phone number for the organization where this application or proceeding is assigned is (571) 272-8300.

31. Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at (866) 217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call (800) 786-9199 (in USA or CANADA) or (571) 272-1000.

/Lawrence-Linh T Nguyen/

Examiner, Art Unit 2897

/Fernando L. Toledo/

Supervisory Patent Examiner, Art
Unit 2897

Notice of References Cited	Application/Control No. 14/729,298		Applicant(s)/Patent Under Reexamination ALPTEKIN ET AL.	
	Examiner LAWRENCE-LINH T. NGUYEN		Art Unit 2897	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-2012/0161324 A1	06-2012	Heinrich; Jens	H01L21/28518	257/757
*	B	US-2014/0027883 A1	01-2014	Hu; Yongjun Jeff	H01L21/743	257/534
	C	US-				
	D	US-				
	E	US-				
	F	US-				
	G	US-				
	H	US-				
	I	US-				
	J	US-				
	K	US-				
	L	US-				
	M	US-				

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Search Notes 	Application/Control No. 14729298	Applicant(s)/Patent Under Reexamination ALPTEKIN ET AL.
	Examiner LAWRENCE-LINH T NGUYEN	Art Unit 2897

CPC- SEARCHED		
Symbol	Date	Examiner
H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261.	09/14/2016	LTN

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner

SEARCH NOTES		
Search Notes	Date	Examiner
Each of the CPC classes listed are searched in combination and/or delimited with text. See EAST search for details	09/14/2016	LTN

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner

/LAWRENCE-LINH T NGUYEN/ Examiner.Art Unit 2897	
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Receipt date: 06/03/2015

Doc code: IDS

Doc description: Information Disclosure Statement (IDS) Filed

14729298 - GAI: 2897

Approved for use through 07/31/2012. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		
	Filing Date		
	First Named Inventor	Emre Alptekin et al.	
	Art Unit		
	Examiner Name		
	Attorney Docket Number	FIS920150066US1	

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GF-ITC-00004889

Receipt date: 06/03/2015 INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		14729298 - GAU: 2897
	Filing Date		
	First Named Inventor	Emre Alptekin et al.	
	Art Unit		
	Examiner Name		
	Attorney Docket Number	FIS920150066US1	

1	Filed Application titled "Metal Semiconductor Alloy Contact Resistance Improvement," US Application No. 14/172,525 filed February 4, 2014, Attorney Docket No. FIS920130170US1, 37 pages.	☐
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EXAMINER SIGNATURE

Examiner Signature	/Lawrence-Linh Nguyen/	Date Considered	09/13/2016
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¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

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GF-ITC-00004890

Receipt date: 06/03/2015 INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		14729298 - GAU: 2897
	Filing Date		
	First Named Inventor	Emre Alptekin et al.	
	Art Unit		
	Examiner Name		
	Attorney Docket Number	FIS920150066US1	

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Amanda J. Smith/	Date (YYYY-MM-DD)	2015-06-03
Name/Print	Amanda J. Smith	Registration Number	71,112

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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GF-ITC-00004891

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GF-ITC-00004892

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
		"1817" and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:42
S1	0	(14/729298).APP.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2016/09/12 20:02
S2	48	((Nicolas) near2 (Breil)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:02
S3	367	((Christian) near2 (Lavoie)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:02
S4	104	((Ahmet) near2 (Ozcan)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:03
S5	133	((Kathryn) near2 (Schonenberg)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:03
S6	104	((Emre) near2 (Alptekin)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:21
S7	9310	("GLOBALFOUNDRIES INC."	US-PGPUB;	OR	ON	2016/09/12

		GLOBALFOUNDRIES).AS.	USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			20:22
S8	55	("8278200").PN. ("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 21:07
S9	181	"7229920" "8450807" "7795124" "6624489" "7785972" "20110297911" "20120112292" "20120326317" "20070249149" "20110147798" "20110221003" "20130240990" "20130009217" "20090289302" "20140027826" "20110147870" "20120129302" "6756651" "7157300" "7262117" "7566875" "7616904" "8765502"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:21
S10	26	(US-20130240990-\$ or US- 20120187460-\$ or US-20120252198-\$ or US-20110221003-\$ or US-20130009217- \$ or US-20090289302-\$ or US- 20140145319-\$ or US-20140138837-\$ or US-20090302480-\$ or US-20100078771- \$ or US-20050085064-\$ or US- 20060246656-\$ or US-20120132966-\$) .did.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:22
S11	13120	H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:24
S12	23052	S1 S2 S3 S4 S5 S6 S7 S11 S9 S10	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:25
S13	1273	S12 and ((opening trench via hole) with dielectric) and ((liner or barrier) with metal) and (metal with (contact or fill))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:25
S14	3241	S12 and ((Drain Gate Source Spacer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/12 22:25

S15	92	S12 and ((opening or trench or via or hole) with dielectric) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:26
S16	516	S13 S14	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/12 22:27
S17	28	S16 and ((opening or trench or via or hole) with dielectric) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:27
S18	11581	((Drain and Gate and Source and Spacer) with (Trench Open Cavity Contact adj Open Contact Metal ((Metal Contact) adj Liner)) with (Dielectric Isolat\$3 Liner Mask Oxide))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:44
S19	650	S12 and ((Drain and Gate and Source and Spacer) with (Trench Open Cavity Contact adj Open Contact Metal ((Metal Contact) adj Liner)) with (Dielectric Isolat\$3 Liner Mask Oxide))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:45
S20	110	S12 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:47
S21	22	S19 S20	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/12 22:48
S22	170152	((Gate Spacer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/12 22:52
S23	63156	S22 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Isolat\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO;	OR	ON	2016/09/12 22:55

			DERWENT; IBM_TDB			
S24	22129	S22 and ((Fusing Fused Fussion Diffuse Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:58
S25	1183	S22 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:58
S26	2141	S22 and (("W" Tungsten "Ti" Titanium "Ta" Tantalum "Mo" Molybdenum "TiN" Titanium adj Nitride "TaN" Tantalum adj Nitride Titanium-Tungsten Titanium adj Tungsten) with ("W" Tungsten "Co" Colbalt "Cu" Copper and "Al" Aluminum) with ("NiPt" Nickel-Platinum Nickel adj Platinum "Ti/TiN" bilayer Liner/Barrier Liner adj2 Barrier))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 23:09
S27	1	"14729298"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S28	49	((Nicolas) near2 (Breil)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S29	368	((Christian) near2 (Lavoie)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S30	105	((Ahmet) near2 (Ozcan)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S31	134	((Kathryn) near2 (Schonenberg)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S32	105	((Emre) near2 (Alptekin)).INV.	US-PGPUB;	OR	ON	2016/09/13

			USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			11:03
S33	245794	("GLOBALFOUNDRIES INC." GLOBALFOUNDRIES "IBM" International adj Business adj Machines "International Business Machines").AS.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:05
S34	13131	H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:12
S35	276	("8278200").PN. ("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN. "7229920" "8450807" "7795124" "6624489" "7785972" "20110297911" "20120112292" "20120326317" "20070249149" "20110147798" "20110221003" "20130240990" "20130009217" "20090289302" "20140027826" "20110147870" "20120129302" "6756651" "7157300" "7262117" "7566875" "7616904" "8765502" (US-20130240990-\$ or US- 20120187460-\$ or US-20120252198-\$ or US-20110221003-\$ or US-20130009217- \$ or US-20090289302-\$ or US- 20140145319-\$ or US-20140138837-\$ or US-20090302480-\$ or US-20100078771- \$ or US-20050085064-\$ or US- 20060246656-\$ or US-20120132966-\$) .did.("20070059878" "7253071" "8101518" "20070181954" "20070093047" "20020111021" "6346465" "20130119483" "20070134898" "6468901" "20080121950" "20100224938" "20120074501" "20070202692" "8338239").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:13
S36	259027	S27 S28 S29 S30 S31 S32 S33 S34 S35	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:15
S37	10581	S36 and ((Drain Gate Source Spacer))	US-PGPUB; USPAT;	AND	ON	2016/09/13 13:25

			USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			
S38	29272	S36 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Insulat\$3 Isolat\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:25
S39	261	S36 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:28
S40	356	S36 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:28
S41	6783	S36 and ((Fusing Fused Fussion Diffuse Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:29
S42	8119	S37 S38	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 13:29
S43	2155	S37 S38 S41	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 13:29
S44	6	S39 S40	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 13:29
S45	0	S43 S44	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 13:29

S46	43	S43 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:42
S47	9	S43 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:43
S48	1709	S43 and ((silicon "Si" germanium "Ge" silicon adj germanium "SiGe" "Si.sub" adj "Ge.sub" silicon adj carbide "SiC" "III-V") near3 (Substrate submount wafer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 16:49
S49	78	S43 and ((Gate adj Dielectric Gate) adj3 ("HfO2" "HfO.sub.2" "ZrO2" "ZrO.sub.2" "La2O3" "La.sub.2O.sub.3" "Al2O3" "Al.sub.zO.sub.3" "TiO2" "TiO.sub.2" "SrTiO3" "SrTiO.sub.3" "LaAlO3" "LaAlO.sub.3" "Y2O3" "Y.sub.2O.sub.3" "HfOxNy" "HfO.sub.xN.sub.y" "ZrOxNy" "ZrO.sub.xN.sub.y" "La2OxNy" "La.sub.2O.sub.xN.sub.y" "Al2OxNy" "Al.sub.zO.sub.xN.sub.y" "TiOxNy" "TiO.sub.xN.sub.y" "SrTiOxNy" "SrTiO.sub.xN.sub.y" "LaAlOxNy" "LaAlO.sub.xN.sub.y" "Y2OxNy" "Y.sub.2O.sub.xN.sub.y" "SiON" "SiNx" "SiN.sub.x Silicate"))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:11
S50	43406	S36 and ((chemical adj vapor adj deposition "chemical vapor deposition" (CVD) "low-pressure CVD" (LPCVD) "plasma-enhanced CVD" (PECVD) "semi-atmosphere CVD" (SACVD) "high density plasma CVD" "high density plasma" (HDPCVD) "rapid thermal CVD" (RTCVD) "ultra-high vacuum CVD" (UHVCVD) "limited reaction processing CVD" (LRPCVD) "metalorganic CVD" (MOCVD) ((sputtering ion adj beam electron adj beam laser adj assisted physical adj vapor atomic adj layer) adj (Deposit deposition)) "electron beam deposition" "sputtering deposition" "ion beam deposition" "laser assisted deposition" "thermal oxidation" "thermal nitridation spin-on" "physical vapor deposition" (PVD) "atomic layer deposition" (ALD) "chemical oxidation" "molecular beam epitaxy" (MBE) plat\$3 evaporat\$3) ((low-pressure low adj pressure plasma-enhanced plasma adj enhanced semi-atmosphere semi adj atmosphere high adj density adj plasma rapid adj thermal ultra-high adj vacuum ultra adj high adj vacuum limited adj reaction adj	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:30

		processing metalorganic) adj (CVD)))				
S51	1853	S43 and ((chemical adj vapor adj deposition "chemical vapor deposition" (CVD) "low-pressure CVD" (LPCVD) "plasma-enhanced CVD" (PECVD) "semi-atmosphere CVD" (SACVD) "high density plasma CVD" "high density plasma" (HDPCVD) "rapid thermal CVD" (RTCVD) "ultra-high vacuum CVD" (UHVCVD) "limited reaction processing CVD" (LRPCVD) "metalorganic CVD" (MOCVD) ((sputtering ion adj beam electron adj beam laser adj assisted physical adj vapor atomic adj layer) adj (Deposit deposition)) "electron beam deposition" "sputtering deposition" "ion beam deposition" "laser assisted deposition" "thermal oxidation" "thermal nitridation spin-on" "physical vapor deposition" (PVD) "atomic layer deposition" (ALD) "chemical oxidation" "molecular beam epitaxy" (MBE) plat\$3 evaporat\$3) ((low-pressure low adj pressure plasma-enhanced plasma adj enhanced semi-atmosphere semi adj atmosphere high adj density adj plasma rapid adj thermal ultra-high adj vacuum ultra adj high adj vacuum limited adj reaction adj processing metalorganic) adj (CVD)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:31
S52	1481	S48 S51	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 17:32
S53	62	S52 and ((Gate adj Dielectric Gate) adj3 ("HfO2" "HfO.sub.2" "ZrO2" "ZrO.sub.2" "La2O3" "La.sub.2O.sub.3" "AlzO3" "Al.sub.zO.sub.3" "TiO2" "TiO.sub.2" "SrTiO3" "SrTiO.sub.3" "LaAlO3" "LaAlO.sub.3" "Y2O3" "Y.sub.2O.sub.3" "HfOxNy" "HfO.sub.xN.sub.y" "ZrOxNy" "ZrO.sub.xN.sub.y" "La2OxNy" "La.sub.2O.sub.xN.sub.y" "AlzOxNy" "Al.sub.zO.sub.xN.sub.y" "TiOxNy" "TiO.sub.xN.sub.y" "SrTiOxNy" "SrTiO.sub.xN.sub.y" "LaAlOxNy" "LaAlO.sub.xN.sub.y" "Y2OxNy" "Y.sub.2O.sub.xN.sub.y" "SiON" "SiNx" "SiN.sub.x" Silicate))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:32
S54	5	S52 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:32
S55	29	S52 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS;	OR	ON	2016/09/13 17:32

			EPO; JPO; DERWENT; IBM_TDB			
S56	34	S52 and (("W" Tungsten "Ti" Titanium "Ta" Tantalum "Mo" Molybdenum "TiN" Titanium adj Nitride "TaN" Tantalum adj Nitride Titanium-Tungsten Titanium adj Tungsten) with ("W" Tungsten "Co" Colbalt "Cu" Copper and "Al" Aluminum) with ("NiPt" Nickel-Platinum Nickel adj Platinum "Ti/TiN" bilayer Liner/Barrier Liner adj2 Barrier))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:39
S57	1481	S52 and ((Fusing Fused Fussion Diffuse Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:39
S58	1452	S52 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Isolat\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:39
S59	318	S52 and ((Drain and Gate and Source and Spacer) with (Trench Open Cavity Contact adj Open Contact Metal ((Metal Contact) adj Liner)) with (Dielectric Isolat\$3 Liner Mask Oxide))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:40
S60	318	S57 S58 S59	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 17:40

9/ 13/ 2016 9:36:40 PM
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BIB DATA SHEET

CONFIRMATION NO. 2837

SERIAL NUMBER	FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.	
14/729,298	06/03/2015	257	2897	FIS920150066US1	
APPLICANTS GLOBALFOUNDRIES INC., Grand Cayman, CAYMAN ISLANDS; INVENTORS Emre Alptekin, Wappingers Falls, NY; Nicolas L. Breil, San Jose, CA; Christian Lavoie, Pleasantville, NY; Ahmet S. Ozcan, Chappaqua, NY; Kathryn T. Schonenberg, Wappingers Falls, NY; ** CONTINUING DATA ***** ** FOREIGN APPLICATIONS ***** ** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 06/11/2015					
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and /LAWRENCE-LINH T NGUYEN/ Acknowledged Examiner's Signature	☐ Met after Allowance Initials	STATE OR COUNTRY NY	SHEETS DRAWINGS 5	TOTAL CLAIMS 20	INDEPENDENT CLAIMS 3
ADDRESS HOFFMAN WARNICK LLC 540 Broadway 4th Floor ALBANY, NY 12207 UNITED STATES					
TITLE CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME					
FILING FEE RECEIVED 1600	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:		☐ All Fees ☐ 1.16 Fees (Filing) ☐ 1.17 Fees (Processing Ext. of time) ☐ 1.18 Fees (Issue) ☐ Other _____ ☐ Credit		

<i>Index of Claims</i> 	Application/Control No. 14729298	Applicant(s)/Patent Under Reexamination ALPTEKIN ET AL.
	Examiner LAWRENCE-LINH T NGUYEN	Art Unit 2897

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant			☐ CPA			☐ T.D.			☐ R.1.47		
CLAIM		DATE									
Final	Original	09/13/2016									
	1	✓									
	2	✓									
	3	✓									
	4	✓									
	5	✓									
	6	✓									
	7	✓									
	8	✓									
	9	✓									
	10	✓									
	11	✓									
	12	✓									
	13	✓									
	14	✓									
	15	N									
	16	N									
	17	N									
	18	N									
	19	N									
	20	N									

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant:	Alptekin et al.	Conf. No.:	2837
Serial No.:	14/729,298	Art Unit:	2897
Filing Date:	06/03/2015	Examiner:	Nguyen, Lawrence-Lin T.
Title:	Contacts to Semiconductor Substrate and Methods of Forming Same	Docket No.:	FIS920150066US1 (GFFT-0376)

Mail Stop Amendment
Commissioner for Patents
PO Box 1450
Alexandria, VA 22313-1450

AMENDMENT

I. INTRODUCTORY COMMENTS

This paper is being filed in response to the Office Action of September 26, 2016. Please amend the above-referenced patent application as follows:

Electronic Acknowledgement Receipt

EFS ID:	27736255
Application Number:	14729298
International Application Number:	
Confirmation Number:	2837
Title of Invention:	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME
First Named Inventor/Applicant Name:	Emre Alptekin
Customer Number:	133964
Filer:	Amanda J Smith
Filer Authorized By:	
Attorney Docket Number:	FIS920150066US1
Receipt Date:	08-DEC-2016
Filing Date:	03-JUN-2015
Time Stamp:	15:10:36
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		FIS920150066US1_ResponseTo OfficeAction_12-08-16.pdf	59140 7e3b2a103cfaab781b3709f2f9a62ef9fac04824	yes	15

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Applicant Arguments/Remarks Made in an Amendment	9	15
	Specification	6	8
	Claims	2	5
	Amendment/Req. Reconsideration-After Non-Final Reject	1	1

Warnings:

Information:

Total Files Size (in bytes):	59140
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IV. REMARKS

Claims 1-14 are currently pending. In this Amendment, claims 2, 6, and 13 and the Specification have been amended, and claims 15-20 have been cancelled without prejudice. No new matter is added. Reconsideration and allowance are requested in view of the accompanying amendments and the remarks below. Applicant is not conceding in this application that the amended or cancelled claims are not patentable over references cited by the Examiner, as the present claim amendments and cancellation are for facilitating expeditious allowance of the claimed subject matter. Further, Applicant reserves the right to pursue the full scope of the subject matter of the original claims in a subsequent patent application that claims priority to the instant application.

Specification Objections

The Specification has been objected to for alleged informalities. Specifically, the Examiner alleges that characters “110” and “112” have both been used to designate “substrate”. (Office Action, page 2). Applicant has amended the Specification to address the alleged informalities. As such, Applicant respectfully requests withdrawal of the objection to the Specification.

Rejection under 35 U.S.C. § 102(a)(1)

In the Office Action, claims 1-14 are rejected under 35 U.S.C. § 102(a)(1), as being anticipated by Heinrich et al. (US Publication 20120161324A1), hereinafter referred to as “Heinrich”. Applicant respectfully traverses the rejections for at least the reasons that follow.

With respect to claim 1, the Examiner alleges that Heinrich teaches “causing a portion of the first liner layer (205) to diffuse into the upper portion of the semiconductor substrate (201) to form a first intermix region (253) at the upper portion of the semiconductor substrate (201).” (Office Action, page 3). Applicant respectfully disagrees. Heinrich’s discloses “during the process 207, the heavily doped semiconductor region 252 may also be exposed within the contact opening 223, while on the other hand corresponding sidewall spacers 205S may be formed on inner surface areas 223S of the contact opening.” (Heinrich, paragraph [0039]). That is, Heinrich’s liner 205 is removed from an upper surface of the heavily doped semiconductor region 252. Heinrich goes on to disclose that “a layer of a refractory metal species 203 may be formed...[and] a heat treatment or a sequence of heat treatments 204 may be applied so as to initiate a chemical reaction of the layer 203 with any underlying semiconductor materials” to form metal silicide 253. (Heinrich, paragraph [0040]). That is, Heinrich’s metal silicide 253, which is equated to Applicant’s first intermix region, is not formed by causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate as claimed. Rather, Heinrich’s liner 205 is removed prior to the forming of Heinrich’s metal silicide 253, and thus, Heinrich’s metal silicide 253 is formed from Heinrich’s refractory metal species 203 and Heinrich’s highly doped semiconductor region 252 by heat treatment.

In another embodiment, Heinrich discloses “the refractory metal 203 may be deposited directly on the semiconductor layer 205 without a preceding material removal process. Consequently, upon initiating a silicidation process, metal silicide may be efficiently formed with the contact opening 223 and in and above the doped region 252, as is also previously explained, while also a metal silicide may be formed on the layer 205 above the dielectric material 221.” (Heinrich, paragraph [0044]). That is, in this embodiment, Heinrich’s metal silicide 253 is

formed above layer 205. Thus, Heinrich does not disclose “causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate” as claimed.

In light of the foregoing, Applicant respectfully submits that Heinrich fails to disclose each and every feature of Applicant’s claim 1. As such, Applicant respectfully requests withdrawal of the 35 U.S.C. § 102(a)(1) rejection of claim 1.

With respect to claim 8, Applicant submits that claim 8 recites *inter alia*, “causing a portion of the first liner layer to diffuse into the upper portion of each of the source and the drain to form a first intermix region at the upper portion of the each of the source and the drain.” As should be clear from the above discussion relative to claim 1, Heinrich does not disclose at least this feature. As such, Applicant respectfully requests withdrawal of the 35 U.S.C. § 102(a)(1) rejection of claim 8.

Should the Office decide to issue a subsequent action issuing any new rejection of claims 1 and/or 8 in this Application, Applicants request that the subsequent action be made non-final, as the new rejection would not be necessitated by Applicants’ amendments herein. (MPEP 706.07(a), “Under present practice, second or any subsequent actions on the merits shall be final, except where the examiner introduces a new ground of rejection that is neither necessitated by applicant's amendment of the claims nor based on information submitted in an information disclosure statement filed during the period set forth in 37 CFR 1.97(c) with the fee set forth in 37 CFR 1.17(p)”).

Claims 2-5, 7, 9-12, and 14 are allowable due to their dependency from allowable claims 1 and 8, respectively. As such, Applicant respectfully requests withdrawal of the 35 U.S.C. § 102(a)(1) rejection of claims 2-5, 7, 9-12, and 14. With respect to claims 6 and 13, Applicant

submits that the Examiner fails to individually address claims 6 and 13 even though he rejects claims 1-14 under 35 U.S.C. § 102(a)(1). As such, Applicant does not yet bear a burden to address any 35 U.S.C. § 102(a)(1) of claims 6 and 13.

More specifically, with respect to claims 2 and 9, since Heinrich fails to disclose the forming of the first intermix region by causing diffusion of the first liner layer, it stands that Heinrich cannot stand to disclose removing a remaining non-diffused portion of the first liner layer to expose the intermix region as claimed. As previously discussed, the Examiner equates Heinrich's metal silicide 253 to the claimed intermix region and Heinrich's liner 205 to the claimed first liner layer. However, Heinrich's liner 205 is removed prior to the formation of Heinrich's metal silicide 253, so it is unclear how the removal of Heinrich's liner 205 exposes Heinrich's metal silicide 253.

With respect to claims 4, 7, 11, and 14, Applicant submits that while Heinrich discloses "layer 203 may comprise any appropriate metal species, such as nickel, platinum and the like, while in other cases cobalt may be used" (Heinrich, paragraph [0040]), this disclosure cannot stand to teach the process of wherein "the depositing the refractory metal layer includes depositing a *bilayer* of titanium and titanium nitride" as claimed. (*Emphasis added*). Should the Examiner persist in this assertion, Applicant respectfully requests that the Examiner provide support for his conclusion that Heinrich teaches the claimed "depositing of a bilayer".

Rejection under 35 U.S.C. § 103

In the Office Action, claims 6 and 13 are rejected under 35 U.S.C. § 103(a) as being unpatentable over Heinrich, hereinafter referred to as "Heinrich", in view of Hu et al. (US

Publication 20140027883A1), hereinafter referred to as “Hu”. Applicant respectfully traverses the rejection for at least the reasons that follow.

First, Applicant notes that in the rejection of claims 6 and 13 under 35 U.S.C. § 103(a), the Examiner merely reproduces the elements of the claims and provides a blanket allegation that Heinrich and Hu teach these features without pointing to specific teachings of Heinrich and/or Hu for which the Examiner relies on in rejection of the claims.

Under MPEP § 2142, “[t]he examiner bears the initial burden of factually supporting any prima facie conclusion of obviousness. If the examiner does not produce a prima facie case, the applicant is under no obligation to submit evidence of nonobviousness.” “The Federal Circuit has stated that ‘rejections on obviousness cannot be sustained with mere conclusory statements; instead, there must be some articulated reasoning with some rational underpinning to support the legal conclusion of obviousness.’ In re Kahn, 441 F.3d 977, 988, 78 USPQ2d 1329, 1336 (Fed. Cir. 2006). See also KSR, 550 U.S. at ___, 82 USPQ2d at 1396 (quoting Federal Circuit statement with approval).” (MPEP § 2142.) As such, Applicant does not yet bear the burden to address the 35 U.S.C. § 103(a) rejections of claims 6 and 13.

However, in the interest of expeditious prosecution, Applicant submits that neither Heinrich nor Hu teach at least “performing a laser anneal of the opening such that a second intermix region is formed at the upper surface of the semiconductor substrate...[and] the second intermix region is composed of materials of the first intermix region and a material used for the refractory metal layer.” Heinrich only teaches forming one metal silicide region 253 within contact opening 223 which is equated to Applicant’s first intermix region. Heinrich does not teach the forming a second intermix region as claimed. Hu teaches “a metal-insulator-metal

(MIM) capacitor having a bottom electrode formed from at least two portions of a metal nitride material.” (Hu, paragraph [0002]).

V. CONCLUSION

Applicant respectfully submits that the Application as presented is in condition for allowance. Should the Examiner believe that anything further is necessary in order to place the application in better condition for allowance, the Examiner is requested to contact Applicant's undersigned representative at the telephone number listed below.

Respectfully submitted,

/Amanda J. Smith /
Amanda J. Smith
Reg. No. 71,112

Date: December 8, 2016

Hoffman Warnick LLC
540 Broadway, 4th Floor
Albany, New York 12207
Phone: (518) 449-0044
Fax: (518) 449-0047

III. AMENDMENTS TO THE SPECIFICATION

Please replace paragraphs [0011], [0013], and [0014] with the following amended paragraphs [0011], [0013], and [0014], respectively:

[0011] Referring now to FIG. 1, a semiconductor structure 100 is shown including a semiconductor substrate 110, an isolation region 112, and a gate structure 114 within a dielectric layer 120 over substrate 112. Substrate ~~[[112]]~~110 may be employed as a semiconductor layer of a semiconductor-on-insulator substrate (not shown), or a bulk substrate (not shown) as understood in the art. Substrate ~~[[112]]~~110 may include any now known or later developed semiconductor material including but not limited to silicon, germanium, silicon germanium, silicon carbide, and those consisting essentially of one or more III-V compound semiconductors having a composition defined by the formula $Al_{X1}Ga_{X2}In_{X3}As_{Y1}P_{Y2}N_{Y3}Sb_{Y4}$, where $X1$, $X2$, $X3$, $Y1$, $Y2$, $Y3$, and $Y4$ represent relative proportions, each greater than or equal to zero and $X1+X2+X3+Y1+Y2+Y3+Y4=1$ (1 being the total relative mole quantity). Other suitable substrates include II-VI compound semiconductors having a composition $Zn_{A1}Cd_{A2}Se_{B1}Te_{B2}$, where $A1$, $A2$, $B1$, and $B2$ are relative proportions each greater than or equal to zero and $A1+A2+B1+B2=1$ (1 being a total mole quantity). Furthermore, a portion or entire substrate may be strained. Substrate ~~[[112]]~~110 may be doped, undoped or contain doped and undoped regions therein.

[0013] Further, FIG. 1 illustrates that a gate 114 may be formed on substrate ~~[[112]]~~110. Gate structure 114 may include a gate dielectric material 116 and a gate conductor 118. Gate structure

114 may be used to control output current of the semiconducting device through electrical or magnetic fields. Gate dielectric 116 may include, but is not limited to: HfO_2 , ZrO_2 , La_2O_3 , Al_2O_3 , TiO_2 , SrTiO_3 , LaAlO_3 , Y_2O_3 , HfO_xN_y , ZrO_xN_y , $\text{La}_2\text{O}_x\text{N}_y$, $\text{Al}_2\text{O}_x\text{N}_y$, TiO_xN_y , SrTiO_xN_y , LaAlO_xN_y , $\text{Y}_2\text{O}_x\text{N}_y$, SiON , SiN_x , a silicate thereof, and an alloy thereof. Each value of x is independently from 0.5 to 3 and each value of y is independently from 0 to 2. In some embodiments, a multilayered gate dielectric structure comprising different gate dielectric materials, e.g., silicon dioxide, and a dielectric metal oxide can be formed and used as gate dielectric material 116. Gate conductor 118 may include, but is not limited to: doped polysilicon, doped silicon germanium, an elemental metal (e.g., tungsten, titanium, tantalum, aluminum, nickel, ruthenium, palladium and platinum), an alloy of at least two elemental metals, an elemental metal nitride (e.g., tungsten nitride, aluminum nitride, and titanium nitride), or multilayered combinations thereof.

[0014] Gate structure 114 may be formed by depositing gate dielectric 116 on substrate 110. As used herein, “depositing” may include any now known or later developed techniques appropriate for the material to be deposited including but are not limited to, for example: chemical vapor deposition (CVD), low-pressure CVD (LPCVD), plasma-enhanced CVD (PECVD), semi-atmosphere CVD (SACVD) and high density plasma CVD (HDPCVD), rapid thermal CVD (RTCVD), ultra-high vacuum CVD (UHVCVD), limited reaction processing CVD (LRPCVD), metalorganic CVD (MOCVD), sputtering deposition, ion beam deposition, electron beam deposition, laser assisted deposition, thermal oxidation, thermal nitridation, spin-on methods, physical vapor deposition (PVD), atomic layer deposition (ALD), chemical oxidation, molecular beam epitaxy (MBE), plating, evaporation. Additionally, subsequent to the

deposition of gate dielectric 116, gate conductor 118 can be deposited on gate dielectric 116. The precise thickness of gate dielectric 116 and gate conductor 118 may vary widely with the intended application.

II. AMENDMENTS TO THE CLAIMS

The following listing of claims replaces all prior versions, and listings, of claims in the application:

1. (Original) A method for forming a contact in a dielectric layer to a semiconductor substrate, the method comprising:

forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate;

depositing a first liner layer to conformally coat the contact opening;

causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate;

depositing a refractory metal layer over the first intermix region; and

depositing a metal in the contact opening thereby forming the contact.

2. (Currently Amended) The method of claim 1, further comprising:

removing a remaining non-diffused portion of the first liner layer to expose the first intermix region at the upper portion of the semiconductor ~~substrate prior~~ substrate prior to depositing the refractory metal layer.

3. (Original) The method of claim 1, wherein the depositing the first liner layer includes depositing at least one of: nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), and ytterbium (Yb).

4. (Original) The method of claim 1, wherein depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN).

5. (Original) The method of claim 1, wherein depositing the first liner layer includes performing physical vapor deposition.

6. (Currently Amended) The method of claim 1, further comprising:

polishing the metal; and

performing a laser anneal of the opening such that a second intermix region is formed at the upper portion of the semiconductor substrate and ~~the upper surface of semiconductor substrate~~ a bottom surface of the contact, wherein the second intermix region is composed of materials ~~used for~~ of the first intermix region and a material used for the refractory metal layer.

7. (Original) The method of claim 1, wherein the depositing the first liner layer includes depositing nickel platinum (NiPt), the depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN), and the depositing the metal in the opening includes depositing tungsten (W).

8. (Original) A method for forming a set of contacts in a dielectric layer to a field effect transistor, the field effect transistor including a source, a drain, and a gate between the source and the drain, the method comprising:

forming a contact opening in the dielectric layer to expose an upper -portion of each of the source and the drain;

depositing a first liner layer to conformally coat each contact opening and causing a portion of the first liner layer to diffuse into the upper portion of each of the source and the drain to form a first intermix region at the upper portion of the each of the source and the drain;

depositing a refractory metal layer over the first intermix region; and

depositing a metal to substantially coat the refractory metal layer and substantially fill each of the openings thereby forming the set of contacts.

9. (Original) The method of claim 8, further comprising:

removing a remaining non-diffused portion of the first liner layer to expose the intermix region prior to depositing the refractory metal layer.

10. (Original) The method of claim 8, wherein the depositing the first liner layer includes depositing nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), and ytterbium (Yb)..

11. (Original) The method of claim 8, wherein the depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN).

12. (Original) The method of claim 8, wherein the depositing the first liner layer includes performing physical vapor deposition.

13. (Currently Amended) The method of claim 8, further comprising:

polishing the metal in each of the openings; and

performing a laser anneal of each of the openings such that a second intermix region is formed at the upper portion of the respective source and drain thereunder, wherein the second intermix region is composed of materials ~~used for~~ of the first intermix region and a material used for the refractory metal layer.

14. (Original) The method of claim 13, wherein the depositing the first liner layer includes depositing nickel platinum (NiPt), the depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN), and the depositing the metal in the opening includes depositing tungsten (W).

15-20. (Cancelled)

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 14/729,298		Filing Date 06/03/2015		☐ To be Mailed	
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO									
APPLICATION AS FILED – PART I									
(Column 1)		(Column 2)							
FOR	NUMBER FILED	NUMBER EXTRA			RATE (\$)	FEE (\$)			
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A			N/A				
☐ SEARCH FEE (37 CFR 1.16(k), (j), or (m))	N/A	N/A			N/A				
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A			N/A				
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*			X \$	=			
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*			X \$	=			
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).								
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))									
* If the difference in column 1 is less than zero, enter "0" in column 2.					TOTAL				
APPLICATION AS AMENDED – PART II									
(Column 1)		(Column 2)		(Column 3)					
AMENDMENT	12/09/2016	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 14	Minus	** 20	= 0	X \$80 =		0	
	Independent (37 CFR 1.16(h))	* 2	Minus	***3	= 0	X \$420 =		0	
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
					TOTAL ADD'L FEE		0		
(Column 1)		(Column 2)		(Column 3)					
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=	X \$		=	
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$		=	
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
					TOTAL ADD'L FEE				
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.									

LIE
STELLA LITTLE

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/729,298	06/03/2015	Emre Alptekin	FIS920150066US1	2837

133964 7590 05/08/2017
HOFFMAN WARNICK LLC
540 Broadway
4th Floor
ALBANY, NY 12207

EXAMINER

NGUYEN, LAWRENCE-LIN T

ART UNIT	PAPER NUMBER
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2897

NOTIFICATION DATE	DELIVERY MODE
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05/08/2017

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

PTOCommunications@hoffmanwarnick.com

Office Action Summary	Application No. 14/729,298	Applicant(s) ALPTEKIN ET AL.	
	Examiner LAWRENCE-LINH T. NGUYEN	Art Unit 2897	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1) ☒ Responsive to communication(s) filed on 12/08/2016.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a) ☒ This action is **FINAL**. 2b) ☐ This action is non-final.

3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☒ Claim(s) 1-20 is/are pending in the application.
5a) Of the above claim(s) 15-20 is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☒ Claim(s) 1-14 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

10) ☐ The specification is objected to by the Examiner.

11) ☐ The drawing(s) filed on _____ is/are: a) ☐ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some** c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.
2. ☐ Certified copies of the priority documents have been received in Application No. _____.
3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☒ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. _____.

4) ☐ Other: _____.

Detailed Action

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

2. This **Final Office** action is responsive to Applicants' Amendment/Req.

Reconsideration-After-Non-Final Rejection filed on **12/08/2016**.

a. In response to the Applicants' request, claims **1-20** are presented for examination: Claims **15-20** are canceled and claims **2, 6, & 13** are amended.

The amended claims are supported by the original disclosure of the present application and therefore, no new matter/issue is raised.

b. Now claims **1-14** are pending.

3. For applicant's benefit portions of the cited reference(s) have been cited to aid in the review of the rejection(s). While every attempt has been made to be thorough and consistent within the rejection it is noted that the *PRIOR ART MUST BE CONSIDERED IN ITS ENTIRETY, INCLUDING DISCLOSURES THAT TEACH AWAY FROM THE CLAIMS*. See MPEP 2141.02 VI.

Specification Objections

4. In the Specification objection to the reference characters "**110**" and "**112**" is withdrawn.

Claim Rejections - 35 USC § 102(a)(1)

5. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

(a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale or otherwise available to the public before the effective filing date of the claimed invention.

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

6. Claims **1-14** are rejected under 35 U.S.C. 102(a1), as being anticipated by

Heinrich et al., (US20120161324A1, hereinafter Heinrich).

7. In reclaim **1**, in FIG. 2, Heinrich discloses a method for forming a contact in a dielectric layer to a semiconductor substrate, the method comprising:

- forming a contact opening(**223**) in the dielectric layer(**221**) over the semiconductor substrate(**201**) to expose an upper portion of the semiconductor substrate(**201**);
 - depositing a first liner layer(**205**) to conformally coat the contact opening(**223**);
 - causing a portion of the first liner layer(**205**) to diffuse into the upper portion of the semiconductor substrate(**201**) to form a first intermix region(**253**) at the upper portion of the semiconductor substrate(**201**);
 - depositing a refractory metal layer(**203**) over the first intermix region;
- and

- depositing a metal(**224a**) in the contact opening(**223**) thereby forming the contact(**223_Contact**) (See § **[0035-0046]**).
8. In reclaim **2**, in FIG. 2b, Heinrich discloses a method of claim **1**, further comprising: removing a remaining non-diffused portion of the first liner layer(**205**) to expose the first intermix region(**223_Bottom**) at the upper portion of the semiconductor substrate(**201**) prior to depositing the refractory metal layer (See § **[0039]**).
9. In reclaim **3**, in FIG. 2, Heinrich discloses a method of claim **1**, wherein the depositing the first liner layer(**205**) includes depositing at least one of: nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), and ytterbium (Yb) (See § **[0006 0009]**).
10. In reclaim **4**, in FIG. 2, Heinrich discloses a method of claim **1**, wherein depositing the refractory metal layer(**203**) includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN) (See § **[0006]**).
11. In reclaim **5**, in FIG. 2, Heinrich discloses a method of claim **1**, wherein depositing the first liner layer includes performing physical vapor deposition (See § **[0006]**).
12. In reclaim **7**, in FIG. 2, Heinrich discloses a method of claim **1**, wherein the depositing the first liner layer(**205**) includes depositing nickel platinum (NiPt), the depositing the refractory metal layer(**203**) includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN), and the depositing the metal(**224a**) in the opening includes depositing tungsten (W) (See § **[0045]**).

13. In reclaim **8**, in FIG. 2, Heinrich discloses a method for forming a set of contacts in a dielectric layer to a field effect transistor, the field effect transistor including a source, a drain, and a gate between the source and the drain, the method comprising:

- forming a contact opening(**223**) in the dielectric layer(**221**) to expose an upper-portion of each of the source(**252_Source**) and the drain(**252_Drain**);
- depositing a first liner layer(**205**) to conformally coat each contact opening(**223**) and causing a portion of the first liner layer(**205**) to diffuse into the upper portion of each of the source(**252_Source**) and the drain(**252_Drain**) to form a first intermix region(**253**) at the upper portion of the each of the source(**252_Source**) and the drain(**252_Drain**);
- depositing a refractory metal layer(**203**) over the first intermix region(**253**); and
- depositing a metal(**224a**) to substantially coat the refractory metal layer(**203**) and substantially fill each of the openings(**223**) thereby forming the set of contacts(**223_Contact**) (*See § [0035-0046]*).

14. In reclaim **9**, in FIG. 2b, Heinrich discloses a method of claim 8, further comprising: removing a remaining non-diffused portion of the first liner layer(**205**) to expose the intermix region(**223_Bottom**) prior to depositing the refractory metal layer(**203**) (*See § [0039]*).

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15. In reclaim **10**, in FIG. 2, Heinrich discloses a method of claim 8, wherein the depositing the first liner layer(**205**) includes depositing nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), and ytterbium (Yb) (*See § [0006 0009]*).

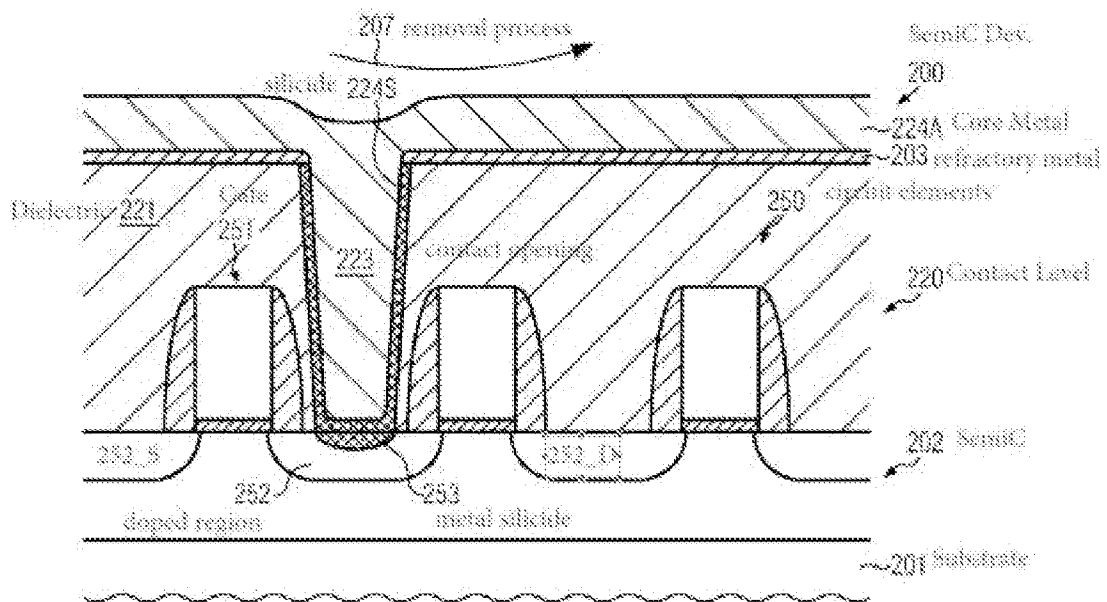
16. In reclaim **11**, in FIG. 2, Heinrich discloses a method of claim 8, wherein the depositing the refractory metal layer(**203**) includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN) (*See § [0006]*).

17. In reclaim **12**, in FIG. 2, Heinrich discloses a method of claim 8, wherein the depositing the first liner layer(**205**) includes performing physical vapor deposition (*See § [0006]*).

18. In reclaim **14**, in FIG. 2, Heinrich discloses a method of claim 13, wherein the depositing the first liner layer(**205**) includes depositing nickel platinum (NiPt), the depositing the refractory metal layer(**203**) includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN), and the depositing the metal(**224a**) in the opening includes depositing tungsten (W) (*See § [0045]*).

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2. (Also see US20120161324A1_FIG2hMod which is reproduced below for convenience).



Claim Rejections - 35 USC § 103

19. The following is a quotation of 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

A patent may not be obtained through the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries set forth in Graham v. John Deere Co., 383 U.S. 1, 148 USPQ 459 (1966), that are applied for establishing a background for determining obviousness under 35 U.S.C. 103(a) are summarized as follows: (See MPEP Ch. 2141)

Determining the scope and contents of the prior art;

Ascertaining the differences between the prior art and the claims in issue; Resolving the level of ordinary skill in the pertinent art; and Evaluating evidence of secondary considerations for indicating obviousness or nonobviousness

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

20. Claims **6 & 13** is rejected under 35 U.S.C. 103(a) as being unpatentable by

Heinrich et al., (**US20120161324A1**, hereinafter **Heinrich**) in view of **Hu et al.**,

(**US20140027883A1**, hereinafter **Hu**).

21. In reclaim **6**, Heinrich is silent in the teaching of a method of claim 1, further comprising: polishing the metal; and performing a laser anneal of the opening such that a second intermix region is formed at the upper portion of the semiconductor substrate and the upper surface of semiconductor substrate, wherein the second

intermix region is composed of materials used for the first intermix region and a material used for the refractory metal layer.

22. But in the same field of endeavor of invention, Hu teaches a method of claim 1, further comprising: polishing the metal; and performing a laser anneal of the opening such that a second intermix region is formed at the upper portion of the semiconductor substrate and a bottom surface of the contact, wherein the second intermix region is composed of materials of the first intermix region and a material used for the refractory metal layer.

23. Hu is evidence that ordinary workers skilled in the art would find reasons, suggestions or motivations to modify the device of Hu with Heinrich. Therefore, at the time the invention was made, it would have been obvious to use the teaching of Hu in the device of Heinrich since the superlattice formed from the first refractory metal nitride material and the second refractory metal nitride material may also improve the thermal stability of the bottom electrode or interconnect (*See § [0016]*).

24. In reclaim **13**, in FIG. 2, Heinrich in view of Hu, Hu teaches a method of claim 8, further comprising: polishing the metal in each of the openings; and performing a laser anneal of each of the openings such that a second intermix region is formed at the upper portion of the respective source and drain thereunder, wherein the second intermix region is composed of materials of the first intermix region and a material used for the refractory metal layer (*See § [0016 0027]*).

Response to Applicant's Remarks

25. With respect to the claims, the following are addressed:
26. Applicant's arguments filed with respect to the claims **1-14**, under USC 102/103 have been fully considered and they are addressed in this Final office action.

Conclusion

27. Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

28. A shortened statutory period for reply to this final action is set to expire THREE(3) MONTHS from the mailing date of this action. In the event a first reply is filed within TWO(2) MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE(3)-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX(6) MONTHS from the mailing date of this final action.

29. Any inquiry concerning this communication or earlier communications from the examiner should be directed to **LAWRENCE-LINH T. NGUYEN** (Email address **Lawrence-Linh.Nguyen@uspto.gov**) whose telephone number is (571) 272-6267.

The examiner can be reached on M-F (8AM - 4PM EST).

30. If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Fernando Toledo can be reached on (571) 272-1867. The fax phone number for the organization where this application or proceeding is assigned is (571) 272-8300.

Art Unit: 2897

31. Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at (866) 217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call (800) 786-9199 (in USA or CANADA) or (571) 272-1000.

/Lawrence-Linh T Nguyen/

Examiner, Art Unit 2897

/Fernando L. Toledo/

Supervisory Patent Examiner, Art
Unit 2897

Notice of References Cited	Application/Control No. 14/729,298		Applicant(s)/Patent Under Reexamination ALPTEKIN ET AL.	
	Examiner LAWRENCE-LINH T. NGUYEN		Art Unit 2897	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-2012/0161324 A1	06-2012	Heinrich; Jens	H01L21/28518	257/757
*	B	US-2014/0027883 A1	01-2014	Hu; Yongjun Jeff	H01L21/743	257/534
	C	US-				
	D	US-				
	E	US-				
	F	US-				
	G	US-				
	H	US-				
	I	US-				
	J	US-				
	K	US-				
	L	US-				
	M	US-				

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

<i>Applicant-Initiated Interview Summary</i>	Application No. 14/729,298	Applicant(s) ALPTEKIN ET AL.	
	Examiner LAWRENCE-LINH T. NGUYEN	Art Unit 2897	

All participants (applicant, applicant's representative, PTO personnel):

(1) LAWRENCE-LINH T. NGUYEN. (3) _____.

(2) Amanda Smith. (4) _____.

Date of Interview: 07/07/2017.

Type: ☒ Telephonic ☐ Video Conference
 ☐ Personal [copy given to: ☐ applicant ☐ applicant's representative]

Exhibit shown or demonstration conducted: ☐ Yes ☒ No.
 If Yes, brief description: _____.

Issues Discussed ☐101 ☐112 ☒102 ☐103 ☐Others
 (For each of the checked box(es) above, please describe below the issue and detailed description of the discussion)

Claim(s) discussed: 1.

Identification of prior art discussed: .

Substance of Interview
 (For each issue discussed, provide a detailed description and indicate if agreement was reached. Some topics may include: identification or clarification of a reference or a portion thereof, claim interpretation, proposed amendments, arguments of any applied references etc....)

Hienrich et al., failed to teach depositing a refractory metal layer over the first intermix region of claim 1. Further search will required.

Applicant recordation instructions: The formal written reply to the last Office action must include the substance of the interview. (See MPEP section 713.04). If a reply to the last Office action has already been filed, applicant is given a non-extendable period of the longer of one month or thirty days from this interview date, or the mailing date of this interview summary form, whichever is later, to file a statement of the substance of the interview

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

☒ Attachment

/LAWRENCE-LINH T NGUYEN/ Examiner, Art Unit 2897	
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Summary of Record of Interview Requirements

Manual of Patent Examining Procedure (MPEP), Section 713.04, Substance of Interview Must be Made of Record

A complete written statement as to the substance of any face-to-face, video conference, or telephone interview with regard to an application must be made of record in the application whether or not an agreement with the examiner was reached at the interview.

Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews

Paragraph (b)

In every instance where reconsideration is requested in view of an interview with an examiner, a complete written statement of the reasons presented at the interview as warranting favorable action must be filed by the applicant. An interview does not remove the necessity for reply to Office action as specified in §§ 1.111, 1.135. (35 U.S.C. 132)

37 CFR §1.2 Business to be transacted in writing.

All business with the Patent or Trademark Office should be transacted in writing. The personal attendance of applicants or their attorneys or agents at the Patent and Trademark Office is unnecessary. The action of the Patent and Trademark Office will be based exclusively on the written record in the Office. No attention will be paid to any alleged oral promise, stipulation, or understanding in relation to which there is disagreement or doubt.

The action of the Patent and Trademark Office cannot be based exclusively on the written record in the Office if that record is itself incomplete through the failure to record the substance of interviews.

It is the responsibility of the applicant or the attorney or agent to make the substance of an interview of record in the application file, unless the examiner indicates he or she will do so. It is the examiner's responsibility to see that such a record is made and to correct material inaccuracies which bear directly on the question of patentability.

Examiners must complete an Interview Summary Form for each interview held where a matter of substance has been discussed during the interview by checking the appropriate boxes and filling in the blanks. Discussions regarding only procedural matters, directed solely to restriction requirements for which interview recordation is otherwise provided for in Section 812.01 of the Manual of Patent Examining Procedure, or pointing out typographical errors or unreadable script in Office actions or the like, are excluded from the interview recordation procedures below. Where the substance of an interview is completely recorded in an Examiners Amendment, no separate Interview Summary Record is required.

The Interview Summary Form shall be given an appropriate Paper No., placed in the right hand portion of the file, and listed on the "Contents" section of the file wrapper. In a personal interview, a duplicate of the Form is given to the applicant (or attorney or agent) at the conclusion of the interview. In the case of a telephone or video-conference interview, the copy is mailed to the applicant's correspondence address either with or prior to the next official communication. If additional correspondence from the examiner is not likely before an allowance or if other circumstances dictate, the Form should be mailed promptly after the interview rather than with the next official communication.

The Form provides for recordation of the following information:

- Application Number (Series Code and Serial Number)
- Name of applicant
- Name of examiner
- Date of interview
- Type of interview (telephonic, video-conference, or personal)
- Name of participant(s) (applicant, attorney or agent, examiner, other PTO personnel, etc.)
- An indication whether or not an exhibit was shown or a demonstration conducted
- An identification of the specific prior art discussed
- An indication whether an agreement was reached and if so, a description of the general nature of the agreement (may be by attachment of a copy of amendments or claims agreed as being allowable). Note: Agreement as to allowability is tentative and does not restrict further action by the examiner to the contrary.
- The signature of the examiner who conducted the interview (if Form is not an attachment to a signed Office action)

It is desirable that the examiner orally remind the applicant of his or her obligation to record the substance of the interview of each case. It should be noted, however, that the Interview Summary Form will not normally be considered a complete and proper recordation of the interview unless it includes, or is supplemented by the applicant or the examiner to include, all of the applicable items required below concerning the substance of the interview.

A complete and proper recordation of the substance of any interview should include at least the following applicable items:

- 1) A brief description of the nature of any exhibit shown or any demonstration conducted,
- 2) an identification of the claims discussed,
- 3) an identification of the specific prior art discussed,
- 4) an identification of the principal proposed amendments of a substantive nature discussed, unless these are already described on the Interview Summary Form completed by the Examiner,
- 5) a brief identification of the general thrust of the principal arguments presented to the examiner,
(The identification of arguments need not be lengthy or elaborate. A verbatim or highly detailed description of the arguments is not required. The identification of the arguments is sufficient if the general nature or thrust of the principal arguments made to the examiner can be understood in the context of the application file. Of course, the applicant may desire to emphasize and fully describe those arguments which he or she feels were or might be persuasive to the examiner.)
- 6) a general indication of any other pertinent matters discussed, and
- 7) if appropriate, the general results or outcome of the interview unless already described in the Interview Summary Form completed by the examiner.

Examiners are expected to carefully review the applicant's record of the substance of an interview. If the record is not complete and accurate, the examiner will give the applicant an extendable one month time period to correct the record.

Examiner to Check for Accuracy

If the claims are allowable for other reasons of record, the examiner should send a letter setting forth the examiner's version of the statement attributed to him or her. If the record is complete and accurate, the examiner should place the indication, "Interview Record OK" on the paper recording the substance of the interview along with the date and the examiner's initials.

Notice of Allowability	Application No. 14/729,298	Applicant(s) ALPTEKIN ET AL.	
	Examiner LAWRENCE-LINH T. NGUYEN	Art Unit 2897	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 07/07/2017.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 1-14. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
 1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has **THREE MONTHS FROM THE "MAILING DATE"** of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|--|--|
| 1. ☒ Notice of References Cited (PTO-892) | 5. ☐ Examiner's Amendment/Comment |
| 2. ☐ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date _____ | 6. ☒ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material | 7. ☐ Other _____. |
| 4. ☒ Interview Summary (PTO-413),
Paper No./Mail Date <u>20170630</u> . | |

/LAWRENCE-LINH T NGUYEN/
Examiner, Art Unit 2897

DETAIL ACTION

Allowable Subject Matter

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.
2. Applicant's arguments, see Response After Final, filed **07/07/2017**, with respect to independent claims **1 & 8** have been fully considered and are persuasive. The **102/103** rejection of independent claims **1 & 8** are withdrawn.
3. The following is an examiner's statement of reasons for allowance
 - a. With respect to independent claim **1**, the prior art teaches:
 - A method for forming a contact in a dielectric layer to a semiconductor substrate, the method comprising: forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate; depositing a first liner layer to conformally coat the contact opening; depositing a refractory metal layer over the first intermix region; and depositing a metal in the contact opening thereby forming the contact..
 - But the prior arts of record do not teach or fairly suggest a limitation of: causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate
 - *Since the Unreacted* (non-diffused) portions of the first liner layer may be removed prior to completion of fabricating the contact. Therefore,

aspects of the present invention do not increase the deposited metal thickness in the contact openings. Thus, contact opening fill issues are avoided and overall contact resistance is improved. There is no reason apparent to the examiner that all the features claimed in claim 1 would be obvious over the prior art.

- Dependent claims 2-7 are respectively dependent on currently amended allowable independent claim 1. Therefore, claims 2-7 incorporate the allowable limitations of claim 1. Consequently, claims 2-7 are allowable.

b. With respect to independent claim 8, the prior art teaches:

- A method for forming a set of contacts in a dielectric layer to a field effect transistor, the field effect transistor including a source, a drain, and a gate between the source and the drain, the method comprising:
forming a contact opening in the dielectric layer to expose an upper -
portion of each of the source and the drain; depositing a refractory metal layer over the first intermix region; and depositing a metal to substantially coat the refractory metal layer and substantially fill each of the openings thereby forming the set of contacts..
- But the prior arts of record do not teach or fairly suggest a limitation of: depositing a first liner layer to conformally coat each contact opening and causing a portion of the first liner layer to diffuse into the upper portion

of each of the source and the drain to form a first intermix region at the upper portion of the each of the source and the drain.

- *Since the Unreacted* (non-diffused) portions of the first liner layer may be removed prior to completion of fabricating the contact. Therefore, aspects of the present invention do not increase the deposited metal thickness in the contact openings. Thus, contact opening fill issues are avoided and overall contact resistance is improved. There is no reason apparent to the examiner that all the features claimed in claim 8 would be obvious over the prior art.

- Dependent claims **9-14** are respectively dependent on currently amended allowable independent claim **8**. Therefore, claims **9-14** incorporate the allowable limitations of claim **8**. Consequently, claims **9-14** are allowable.

c. This application is in condition for allowance except for the presence of claims **15-20** directed to an invention non-elected without traverse. Accordingly, claims **15-20** have been cancelled.

3. Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance".

Conclusion

4. Any inquiry concerning this communication or earlier communications from the examiner should be directed to **LAWRENCE-LINH T. NGUYEN** whose telephone number is **(571) 272-6267** and fax number is **(571) 273-6267**. The examiner can normally be reached on M-F (Flex hours). If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, **Fernando Toledo** can be reached on **571-272-1867**. The fax phone number for the organization where this application or proceeding is assigned is **571-273-8300**.

5. Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/LAWRENCE-LINH T NGUYEN/

Examiner, Art Unit 2897

/Fernando L. Toledo/

Supervisory Patent Examiner, Art Unit 2897

<i>Applicant-Initiated Interview Summary</i>	Application No. 14/729,298	Applicant(s) ALPTEKIN ET AL.	
	Examiner LAWRENCE-LINH T. NGUYEN	Art Unit 2897	

All participants (applicant, applicant's representative, PTO personnel):

(1) LAWRENCE-LINH T. NGUYEN. (3) ____.

(2) Amanda Smith. (4) ____.

Date of Interview: 07/07/2017.

Type: ☒ Telephonic ☐ Video Conference
 ☐ Personal [copy given to: ☐ applicant ☐ applicant's representative]

Exhibit shown or demonstration conducted: ☐ Yes ☒ No.
 If Yes, brief description: ____.

Issues Discussed ☐101 ☐112 ☒102 ☐103 ☐Others
 (For each of the checked box(es) above, please describe below the issue and detailed description of the discussion)

Claim(s) discussed: 1.

Identification of prior art discussed: .

Substance of Interview
 (For each issue discussed, provide a detailed description and indicate if agreement was reached. Some topics may include: identification or clarification of a reference or a portion thereof, claim interpretation, proposed amendments, arguments of any applied references etc...)

Hienrich et al., failed to teach depositing a refractory metal layer over the first intermix region of claim 1. Further search will required.

Applicant recordation instructions: The formal written reply to the last Office action must include the substance of the interview. (See MPEP section 713.04). If a reply to the last Office action has already been filed, applicant is given a non-extendable period of the longer of one month or thirty days from this interview date, or the mailing date of this interview summary form, whichever is later, to file a statement of the substance of the interview

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

☒ Attachment

/LAWRENCE-LINH T NGUYEN/ Examiner, Art Unit 2897	
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Summary of Record of Interview Requirements

Manual of Patent Examining Procedure (MPEP), Section 713.04, Substance of Interview Must be Made of Record

A complete written statement as to the substance of any face-to-face, video conference, or telephone interview with regard to an application must be made of record in the application whether or not an agreement with the examiner was reached at the interview.

Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews

Paragraph (b)

In every instance where reconsideration is requested in view of an interview with an examiner, a complete written statement of the reasons presented at the interview as warranting favorable action must be filed by the applicant. An interview does not remove the necessity for reply to Office action as specified in §§ 1.111, 1.135. (35 U.S.C. 132)

37 CFR §1.2 Business to be transacted in writing.

All business with the Patent or Trademark Office should be transacted in writing. The personal attendance of applicants or their attorneys or agents at the Patent and Trademark Office is unnecessary. The action of the Patent and Trademark Office will be based exclusively on the written record in the Office. No attention will be paid to any alleged oral promise, stipulation, or understanding in relation to which there is disagreement or doubt.

The action of the Patent and Trademark Office cannot be based exclusively on the written record in the Office if that record is itself incomplete through the failure to record the substance of interviews.

It is the responsibility of the applicant or the attorney or agent to make the substance of an interview of record in the application file, unless the examiner indicates he or she will do so. It is the examiner's responsibility to see that such a record is made and to correct material inaccuracies which bear directly on the question of patentability.

Examiners must complete an Interview Summary Form for each interview held where a matter of substance has been discussed during the interview by checking the appropriate boxes and filling in the blanks. Discussions regarding only procedural matters, directed solely to restriction requirements for which interview recordation is otherwise provided for in Section 812.01 of the Manual of Patent Examining Procedure, or pointing out typographical errors or unreadable script in Office actions or the like, are excluded from the interview recordation procedures below. Where the substance of an interview is completely recorded in an Examiners Amendment, no separate Interview Summary Record is required.

The Interview Summary Form shall be given an appropriate Paper No., placed in the right hand portion of the file, and listed on the "Contents" section of the file wrapper. In a personal interview, a duplicate of the Form is given to the applicant (or attorney or agent) at the conclusion of the interview. In the case of a telephone or video-conference interview, the copy is mailed to the applicant's correspondence address either with or prior to the next official communication. If additional correspondence from the examiner is not likely before an allowance or if other circumstances dictate, the Form should be mailed promptly after the interview rather than with the next official communication.

The Form provides for recordation of the following information:

- Application Number (Series Code and Serial Number)
- Name of applicant
- Name of examiner
- Date of interview
- Type of interview (telephonic, video-conference, or personal)
- Name of participant(s) (applicant, attorney or agent, examiner, other PTO personnel, etc.)
- An indication whether or not an exhibit was shown or a demonstration conducted
- An identification of the specific prior art discussed
- An indication whether an agreement was reached and if so, a description of the general nature of the agreement (may be by attachment of a copy of amendments or claims agreed as being allowable). Note: Agreement as to allowability is tentative and does not restrict further action by the examiner to the contrary.
- The signature of the examiner who conducted the interview (if Form is not an attachment to a signed Office action)

It is desirable that the examiner orally remind the applicant of his or her obligation to record the substance of the interview of each case. It should be noted, however, that the Interview Summary Form will not normally be considered a complete and proper recordation of the interview unless it includes, or is supplemented by the applicant or the examiner to include, all of the applicable items required below concerning the substance of the interview.

A complete and proper recordation of the substance of any interview should include at least the following applicable items:

- 1) A brief description of the nature of any exhibit shown or any demonstration conducted,
- 2) an identification of the claims discussed,
- 3) an identification of the specific prior art discussed,
- 4) an identification of the principal proposed amendments of a substantive nature discussed, unless these are already described on the Interview Summary Form completed by the Examiner,
- 5) a brief identification of the general thrust of the principal arguments presented to the examiner,
(The identification of arguments need not be lengthy or elaborate. A verbatim or highly detailed description of the arguments is not required. The identification of the arguments is sufficient if the general nature or thrust of the principal arguments made to the examiner can be understood in the context of the application file. Of course, the applicant may desire to emphasize and fully describe those arguments which he or she feels were or might be persuasive to the examiner.)
- 6) a general indication of any other pertinent matters discussed, and
- 7) if appropriate, the general results or outcome of the interview unless already described in the Interview Summary Form completed by the examiner.

Examiners are expected to carefully review the applicant's record of the substance of an interview. If the record is not complete and accurate, the examiner will give the applicant an extendable one month time period to correct the record.

Examiner to Check for Accuracy

If the claims are allowable for other reasons of record, the examiner should send a letter setting forth the examiner's version of the statement attributed to him or her. If the record is complete and accurate, the examiner should place the indication, "Interview Record OK" on the paper recording the substance of the interview along with the date and the examiner's initials.

Notice of References Cited	Application/Control No. 14/729,298		Applicant(s)/Patent Under Reexamination ALPTEKIN ET AL.	
	Examiner LAWRENCE-LINH T. NGUYEN		Art Unit 2897	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-2012/0161324 A1	06-2012	Heinrich; Jens	H01L21/28518	257/757
*	B	US-2014/0027883 A1	01-2014	Hu; Yongjun Jeff	H01L21/743	257/534
	C	US-				
	D	US-				
	E	US-				
	F	US-				
	G	US-				
	H	US-				
	I	US-				
	J	US-				
	K	US-				
	L	US-				
	M	US-				

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Search Notes 	Application/Control No. 14729298	Applicant(s)/Patent Under Reexamination ALPTEKIN ET AL.
	Examiner LAWRENCE-LINH T NGUYEN	Art Unit 2897

CPC- SEARCHED		
Symbol	Date	Examiner
H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261.	09/14/2016	LTN

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

SEARCH NOTES		
Search Notes	Date	Examiner
Each of the CPC classes listed are searched in combination and/or delimited with text. See EAST search for details	09/14/2016	LTN

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner
	Each of the CPC classes listed are searched in combination and/or delimited with text. See EAST search for details	09/14/2017	LTN

/LAWRENCE-LINH T NGUYEN/ Examiner.Art Unit 2897	
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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant(s): Emre Alptekin

Conf. No.: 2837

Serial No.: 14/729,298

Art. Unit: 2897

Filed: June 3, 2015

Examiner: Nguyen, Lawrence –Lin T.

Docket No.: FIS920150066US1
(GFFT-0376-US)

Title: **CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME**

Examiner Nguyen, Lawrence –Lin T.
By Email: Lawrence-Linh.Nguyen@USPTO.GOV

**AGENDA FOR DISCUSSION IN TELEPHONE INTERVIEW
WITH EXAMINER**

Sir:

I. INTRODUCTORY COMMENTS

Applicant gratefully acknowledges the Examiner's courtesy in extending the opportunity to discuss the above-captioned patent application in a telephone interview scheduled for Friday, June 30, 2017 at 2:00 PM EDT. The proposed interview participants are to include Examiner Nguyen and Applicant's undersigned representative, Amanda Smith (Reg. No. 71,112).

II. AGENDA- DO NOT ENTER

A. Applicant would like to discuss the 35 U.S.C. § 102(a)(1) rejection of claim 1. Specifically, Applicant submits that, in the instant Office Action, the Examiner has failed to address Applicant's arguments submitted in the December 8, 2016 response. In the instant Office Action, the rejection of claim 1 from the Office Action dated September 26, 2016 is merely reproduced without any response to Applicant's previously submitted arguments. Applicant requests clarification on the rejections and why the Examiner did not find Applicant's arguments persuasive so that Applicant can better advance prosecution.

B. In the alternative, Applicant also proposes the following amendment to claim 1 to discuss relative to the 35 U.S.C. § 102(a)(1) rejection of claim 1.

1. (Proposed Amendment – Do Not Enter) A method for forming a contact in a dielectric layer to a semiconductor substrate, the method comprising:

forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate;

depositing a first liner layer to conformally coat the contact opening;

~~causing such that~~ a portion of the first liner layer to ~~diffuse~~ diffuses into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate;

depositing a refractory metal layer over the first intermix region; and

depositing a metal in the contact opening thereby forming the contact.

Applicant proposes this amendment to discuss with the Examiner without prejudice. Applicant does not concede to the rejections in the pending Office Action dated May 8, 2017. Applicant reserves the right to pursue broader scope in a response to the pending Office Action.

C. Applicant would like to discuss the 35 U.S.C. §103 rejection of claim 6. Specifically, Applicant would like to discuss the limitation of “performing a laser anneal of the opening such that a second intermix region is formed at the upper portion of the semiconductor substrate and a bottom surface of the contact, wherein the second intermix region is composed of materials of the first intermix region and a material used for the refractory metal layer” relative to Heinrich and Hu.

III. CONCLUSION

Thank you in advance for your consideration.

Respectfully submitted,

/ Amanda Smith /
Amanda Smith
Reg. No. 71,112

Date: June 29, 2017

Hoffman Warnick LLC
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Phone: (518) 449-0044
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